LEANDRO SERRANO

SOFTWARE DEVELOPER | UTN STUDENT

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GitHub | LinkedIn | Portfolio

PROFESSIONAL PROFILE

Programming student at Universidad Tecnologica Nacional (UTN), focused on software development, with hands-on experience in **C++, Java, and Python**, Object-Oriented Programming, data structures, and databases. Experience building desktop tools, web applications, and Java projects released to real user communities. Currently taking part in a professional internship project with Volkswagen Argentina, applying software development and data analysis in an industrial environment. Seeking an opportunity to grow as a software developer while contributing strong problem-solving skills, self-directed learning, and sound technical judgment.

TECHNICAL SKILLS

Languages: C++, Java, Python, SQL, JavaScript, TypeScript, HTML/CSS; academic experience with C# and .NET.

Development: OOP, STL, data structures, memory management, design patterns, REST APIs, React, Next.js, Node.js/Express, Qt 6, SFML, OpenGL, Dear ImGui.

Data & Tools: PostgreSQL, SQLite/Turso, Pandas, Scikit-learn, Matplotlib, Power BI/DAX, Git, GitHub, CMake, Gradle, Docker, Jira.

AI & Automation: Use of AI coding assistants, LLM APIs, and scripting for prototyping, code review, and development workflow automation.

Languages: Spanish: Native | English: C1 - Advanced.

PROFESSIONAL EXPERIENCE

Professional Internship Project - Volkswagen Argentina / UTN

2025 - 2026

- Contributed with the internship team to the development of a software system supporting quality control for an automotive production line.
- Implemented analysis of torque, pressure, and flow measurements using Statistical Process Control (SPC), Cpk calculations, and linear regression to anticipate process deviations.
- Developed a dashboard for variable monitoring and scenario simulation. Main stack: Node.js, Express, React, Vite, Chart.js, PostgreSQL, and Docker.

Independent Java Mod Developer - Minecraft / Hytale

2019 - Present

- Develop and maintain Java mods published on CurseForge, with more than 300,000 cumulative downloads.
- Use Gradle, dependency management, and modding APIs such as Forge/Fabric; maintain compatibility across versions, resolve bugs, and incorporate improvements based on user feedback.

Freelance Technical Support

Tigre, Buenos Aires | 2020 - Present

- Diagnose and resolve hardware and software issues, build PCs, and provide user support, developing practical troubleshooting and technical support experience.

SELECTED PROJECTS

FastEditor | *C++, OpenGL, Dear ImGui*

- Experimental desktop text editor written in C++, with a Dear ImGui interface and OpenGL-based rendering.
- Implemented a Gap Buffer data structure for incremental text editing, along with keyboard/mouse navigation and multiple-tab handling.
- Includes basic syntax highlighting for C/C++ and light/dark theme support.

GameStorageManager | *C++, CMake, Qt 6*

- Windows desktop utility for applying and reverting NTFS compression on game directories through compact.exe.

- Implements compression-mode selection logic and pre-operation validations before modifying files, aiming to reduce disk usage without altering game content.
- Code organized into UI, application logic, and system-access layers, with analysis, execution, and rollback operations.

TaskFlow - Kanban Manager | *Next.js, TypeScript, Turso, Drizzle ORM*
- Full-stack Kanban application deployed on Vercel, with persistent data, drag-and-drop between columns, subtasks, and form validation.
- Built with Next.js App Router, Server Actions, Drizzle ORM, Tailwind CSS, and Zod.

ISLANDER - 2D Game Engine | *C++, SFML*
- 2D game/engine project featuring a game loop, resource management, tilemaps, collision handling, and smart pointers for safer memory management.

EDUCATION

Universidad Tecnologica Nacional (UTN) - University Technical Degree in Programming
2024 - Present | Buenos Aires
- Coursework in programming logic, Object-Oriented Programming, data structures, software architecture, and databases.
- Academic learning and practice with C++, C#, .NET, and Java, complemented by personal projects and the Professional Internship Project with Volkswagen Argentina.

Fundacion Empujar - UTN Buenos Aires | Tester Master QA
2024
- Software quality assurance training: test case design and execution, bug reporting, documentation, and fundamentals of manual and automated testing.

Codo a Codo Buenos Aires | Full-Stack Python
2023
- Web development with Python/Flask, HTML, CSS, and JavaScript, relational databases, and application deployment fundamentals.

Coder House Online | Data Science & Analytics
2022 - 2023
- Data analysis and visualization with Python, Pandas, and Matplotlib; statistics, introductory Machine Learning with Scikit-learn, and Power BI reporting.

ADDITIONAL PROJECTS & PRACTICE

Data Analysis - Power BI & DAX | *Power BI, DAX*
- Interactive report featuring data modeling, table relationships, DAX measures, KPIs, and visualizations for trend analysis.

UTN-Pathfinder | *React, JavaScript*
- Project developed during a 72-hour hackathon to visualize search algorithms and practice logic, interaction design, and problem solving under time constraints.

Manual Testing Project | *QA*
- Designed and executed test cases, documented defects, and recorded software quality processes.

Predictive Models | *Python, Pandas, Scikit-learn*
- Practice projects covering data cleaning and analysis, regression, KNN, metric evaluation, and result visualization with Matplotlib.

Pomodoro Logger | *React, Vite*
- Minimalist frontend application with a timer, checklist, and state management.

ADDITIONAL INFORMATION

Driver's license: Category B.1 (light vehicles).